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Seat No : _____

No. of Printed Pages : 2

Sardar Patel University
External Examination
M.Sc. (Information Technology) Semester-I
PS01CINT05: Operating System Concepts
23rd November, 2019, Saturday

Time: 10:00 AM to 01:00 PM

Max. Marks: 70

Q-1 Choose the most appropriate option for each question.

[08]

- (i) Which of the following is not a component of operating system?
 - a) System Program
 - b) Application Program
 - c) Loader
 - d) Linker
- (ii) Full form of SMP is _____.
 - a) Symmetric Multiprocessing
 - b) Symmetric Microprocessing
 - c) Synchronous Multiprocessing
 - d) Synchronous Microprocessing
- (iii) _____ Model maps many user level threads to one kernel thread.
 - a) Many to One
 - b) One to Many
 - c) Many to Many
 - d) All of these
- (iv) The operating system starts in _____ mode.
 - a) User
 - b) Kernel
 - c) Supervisor
 - d) Both (b) and (c)
- (v) Which of the following algorithm is not a stack algorithm?
 - a) FIFO
 - b) LRU
 - c) Optimal
 - d) None of these
- (vi) Which of the following structure is used in segmentation?
 - a) Page Table
 - b) Segment Table
 - c) TLB
 - d) None of these
- (vii) Which of the following is not a type of the file?
 - a) Text
 - b) Object
 - c) Audio
 - d) None of these
- (viii) Which of the following algorithm is used for deadlock avoidance?
 - a) Resource Allocation
 - b) Banker's
 - c) Peterson's
 - d) None of these

Q-2 Answer ANY SEVEN from the following

[14]

- (i) What is real time operating system?
- (ii) Differentiate: shared memory v/s message passing systems
- (iii) What is system call? Give any two examples of system calls.
- (iv) What do you mean by non preemptive CPU Scheduling algorithm?
- (v) What is monitor?
- (vi) Briefly explain compile time address binding.
- (vii) What is page fault?
- (viii) What are major benefits of using RAID?
- (ix) Briefly explain any two file operations.

(1)

(P.T.O.)

Q-3

- A. List different structures of operating systems. Explain layered approach of designing operating system along with its advantages and disadvantages. [06]
- B. Write a note on clustered systems. [06]

OR

- B. Write a note on virtual machines.

Q-4

- A. Define Ready Queue and Device Queue. Draw and explain queuing diagram. [06]
- B. What is deadlock? Explain necessary conditions for deadlock. [06]

OR

- B. Explain FCFS CPU scheduling algorithm by giving suitable example of your choice.

Q-5

- A. Explain contiguous memory allocation scheme. Explain internal and external fragmentation by giving suitable examples. [06]
- B. What is page table? List the possible structures of page table and explain any one of them in detail.. [06]

OR

- B. Explain optimal page replacement algorithm by considering following reference string:
7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1
Assume that each process is given three frames in the memory.
Calculate number of page faults for the same.

Q-6

- A. Define the terms "Seek Time", "Disk Bandwidth", and "Rotational Latency" in context of hard disk. Draw the structural diagram of Hard Disk Drive. [06]
- B. What is file? Briefly explain different types of files and their extensions [06]

OR

- B. What do you mean by different levels of RAID? Explain any two levels of RAID.

— X —

(2)